A Negative Answer to Ostrovskii's Projection-Factorization Problem

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Abstract

Ostrovskii asked whether there is a universal constant controlling a projection onto an intermediate finite-dimensional space while requiring its composition with a second projection to be a minimal projection onto a smaller space. We first verify an elementary correction argument that gives the dimension-dependent upper bound $1 + 2\sqrt{d}$, where d is the dimension of the smaller space. We then give a counterexample to the universal-constant question. More precisely, for every $d \geq 2$ we construct a finite-dimensional triple $X_{1,d} \subset X_{2,d} \subset X_{3,d}$ with $\dim X_{1,d} = d$ and $\lambda(X_{2,d}, X_{3,d}) \leq 2$ such that every pair of projections $P_1 : X_{2,d} \to X_{1,d}$ and $P_2 : X_{3,d} \to X_{2,d}$ satisfying $\|P_1P_2\| = \lambda(X_{1,d}, X_{3,d})$ must obey

$$\|P_2\| \geq \sqrt{d/6} - 2.$$

Consequently no universal constant exists. The only non-elementary input is the known unique minimality of the classical Rademacher projection in $L_p[0,1]$, $1 < p < \infty$.

Source problem screenshot

The source question is Problem 3 in M. I. Ostrovskii, *Compositions of projections in Banach spaces and relations between approximation properties*, arXiv:0811.1763.

Remark. By [8, Theorem 5] the sufficient enlargement $A = \lambda(\ell_2^k)B_2^k + \lambda(\ell_2^{n-k})B_2^{n-k}$ is minimal. Hence $\mathrm{cl}(P_2(B_{L_\infty(\mu)})) = A$ and $||P_2|| = ((\lambda(\ell_2^k))^2 + (\lambda(\ell_2^{n-k}))^2)^{1/2}$.

Is it always like this? More precisely

Problem 3 *Does there exists a universal constant $C \in [1, \infty)$ such that for each triple $X_1 \subset X_2 \subset X_3$ of Banach spaces, with X_1 and X_2 finite dimensional, there exist projections $P_1 : X_2 \to X_1$ and $P_2 : X_3 \to X_2$, such that $||P_2|| \leq C\lambda(X_2, X_3)$ and $||P_1P_2|| = \lambda(X_1, X_3)$?*

Another version of this problem (which will be particularly interesting if Problem 3 has a negative answer):

Status and idea

This packet combines two compatible results. First, the correction formula in Section 2 proves a positive dimension-dependent theorem: Problem 3 is always solvable with constant $C_d = 1 + 2\sqrt{d}$ when $d = \dim X_1$. Second, Theorem 1.1 gives a counterexample family to the dimension-free version asked in the source paper. Together they show that the correct worst-case dependence on $\dim X_1$ is of order $\sqrt{d}$, up to absolute constants.

The idea of the negative answer is to start with a large-norm projection that is uniquely minimal. A hyperplane containing its range then forces any factorization whose product remains minimal to pass through a projection with large norm. The required uniquely minimal projections are the finite Rademacher projections in $L_{2d}(\{-1,1\}^d)$, using the known Lewicki–Skrzypek uniqueness theorem for the classical Rademacher projection.

1 The problem and the main result

For a finite-dimensional subspace Y of a Banach space X, write

$$\lambda(Y, X) = \inf\{\|Q\| : Q : X \to Y \text{ is a projection}\}.$$

Ostrovskii's Problem 3 [3] asks whether there is a constant $C < \infty$ such that, for every triple $X_1 \subset X_2 \subset X_3$ with X_1 and X_2 finite-dimensional, one can find projections

$$P_1 : X_2 \to X_1, \qquad P_2 : X_3 \to X_2$$

for which

$$\|P_2\| \le C\lambda(X_2, X_3) \quad \text{and} \quad \|P_1 P_2\| = \lambda(X_1, X_3).$$

All spaces in the counterexample below are real. This is enough to disprove the universal statement.

Theorem 1.1 (Negative answer)**.** *For every integer $d \ge 2$ there are finite-dimensional real Banach spaces*

$$X_{1,d} \subset X_{2,d} \subset X_{3,d}, \qquad \dim X_{1,d} = d,$$

such that $\lambda(X_{2,d}, X_{3,d}) \le 2$ and every pair of projections $P_1 : X_{2,d} \to X_{1,d}$ and $P_2 : X_{3,d} \to X_{2,d}$ satisfying

$$\|P_1 P_2\| = \lambda(X_{1,d}, X_{3,d})$$

necessarily satisfies

$$\|P_2\| \ge \sqrt{d/6} - 2.$$

In particular,

$$\frac{\|P_2\|}{\lambda(X_{2,d}, X_{3,d})} \ge \frac{\sqrt{d/6} - 2}{2},$$

so Ostrovskii's Problem 3 has a negative answer.

The construction has $\dim X_{3,d} = 2^d$ and $\dim X_{2,d} = 2^d - 1$. The proof is given in Sections 3 and 4.

2 Verification of the dimension-dependent correction argument

We first record the partial result from the supplied solution packet. The argument is correct; in fact it gives the slightly sharper unfactored estimate in (1) below.

Proposition 2.1 (Correction formula). *Let $X_1 \subset X_2 \subset X_3$, where X_1 and X_2 are finite-dimensional. There exist projections $P_1 : X_2 \to X_1$ and $P_2 : X_3 \to X_2$ such that*

$$\|P_1 P_2\| = \lambda(X_1, X_3)$$

and

$$\|P_2\| \leq \lambda(X_2, X_3) + \lambda(X_1, X_3) + \lambda(X_1, X_3)\lambda(X_2, X_3). \tag{1}$$

If $X_2 \neq \{0\}$, then in particular

$$\|P_2\| \leq (1 + 2\lambda(X_1, X_3))\lambda(X_2, X_3).$$

Proof. Projection constants onto finite-dimensional ranges are attained. Choose minimal projections

$$P : X_3 \to X_1, \qquad R : X_3 \to X_2.$$

Define

$$P_1 = P\restriction_{X_2}, \qquad P_2 = R + P - PR.$$

The range of P_2 is contained in X_2. If $x \in X_2$, then $Rx = x$ and therefore $P_2 x = x + Px - Px = x$. Thus P_2 is a projection onto X_2. Moreover,

$$P_1 P_2 = P(R + P - PR) = PR + P - P^2 R = P.$$

Hence $P_1 P_2$ is minimal onto X_1. Finally,

$$\|P_2\| \leq \|R\| + \|P\| + \|P\|\,\|R\|,$$

which is (1). Since $\lambda(X_2, X_3) \geq 1$ for a nonzero range, the displayed factored estimate follows. □

For completeness, attainment follows by fixing a basis of the target, writing a bounded minimizing net of projections in terms of coordinate functionals, and taking weak-star convergent subnets of those functionals. The projection identities pass to the limit and the operator norm is lower semicontinuous.

Corollary 2.2 (Fixed-dimensional upper bound). *If $\dim X_1 = d$, then the conclusion requested in Problem 3 holds for that triple with*

$$C_d = 1 + 2\sqrt{d}.$$

Proof. The Kadets–Snobar theorem [1] gives $\lambda(X_1, X_3) \leq \sqrt{d}$; apply Proposition 2.1. □

3 An obstruction from a unique minimal projection

The key observation is abstract. A unique minimal projection with large norm forces a bad intermediate hyperplane.

Lemma 3.1. *Let X be finite-dimensional, let $E \subsetneq X$, and let $P : X \to E$ be a projection. Put $K = \ker P$, let $q : X \to X/E$ be the quotient map, and define*

$$J = q\restriction_K : K \longrightarrow X/E.$$

Then J is an isomorphism and

$$\|J^{-1}\| = \|I - P\|.$$

Consequently, there is a functional $\varphi \in E^\perp$ such that

$$\|\varphi\| = 1, \qquad \|\varphi\restriction_K\| = \frac{1}{\|I - P\|}. \tag{2}$$

Proof. Because $X = E \oplus K$, the map J is bijective and

$$J^{-1}(qx) = (I - P)x.$$

For every $x \in X$ and $e \in E$, $(I - P)(x - e) = (I - P)x$. Taking an infimum over e gives

$$\|(I - P)x\| \le \|I - P\| \, \|qx\|,$$

so $\|J^{-1}\| \le \|I - P\|$. Conversely,

$$\|(I - P)x\| = \|J^{-1}(qx)\| \le \|J^{-1}\| \, \|qx\| \le \|J^{-1}\| \, \|x\|,$$

and equality of the two norms follows.

Since J^* is an isomorphism between finite-dimensional spaces,

$$\min_{\|\psi\|=1} \|J^*\psi\| = \frac{1}{\|(J^*)^{-1}\|} = \frac{1}{\|J^{-1}\|}.$$

Choose a unit vector $\psi \in (X/E)^*$ attaining this minimum and set $\varphi = q^*\psi$. The adjoint q^* is an isometry from $(X/E)^*$ onto $E^\perp$, and $\varphi\restriction_K = J^*\psi$. This proves (2). $\square$

Proposition 3.2 (Bad intermediate hyperplane)**.** *Let X be finite-dimensional, let $E \subsetneq X$, and suppose that $P : X \to E$ is the unique minimal projection. There is a hyperplane $F \subset X$ containing E such that*

$$\lambda(F, X) \le 2$$

and the following holds: whenever $A : F \to E$ and $R : X \to F$ are projections for which AR is minimal onto E, one has

$$\|R\| \ge \|I - P\| - 1 \ge \|P\| - 2. \tag{3}$$

Proof. Let $K = \ker P$, choose φ as in Lemma 3.1, and set $F = \ker\varphi$. Since $\varphi \in E^\perp$, we have $E \subset F$. A norm-one functional on a finite-dimensional space attains its norm, so one can choose $v \in X$ with $\|v\| = 1$ and $\varphi(v) = 1$. Then

$$R_0 = I - v \otimes \varphi$$

is a projection onto F and $\|R_0\| \le 2$. Hence $\lambda(F, X) \le 2$.

Now let A and R be as in the statement. The product AR is a projection onto E. Since it is minimal and P is the unique minimal projection, $AR = P$. Every projection onto the hyperplane $F = \ker\varphi$ has the form

$$R = I - u \otimes \varphi, \qquad \varphi(u) = 1.$$

Because $\ker R \subset \ker(AR) = \ker P = K$, the vector u belongs to K. Thus (2) yields

$$1 = |\varphi(u)| \le \|\varphi\restriction_K\| \, \|u\| = \frac{\|u\|}{\|I - P\|},$$

so $\|u\| \ge \|I - P\|$. Since $\|\varphi\| = 1$,

$$\|I - R\| = \|u \otimes \varphi\| = \|u\| \ge \|I - P\|.$$

The triangle inequality gives $\|R\| \ge \|I - R\| - 1$, proving the first inequality in (3). The second follows from $\|P\| \le 1 + \|I - P\|$. $\square$

4 Unique minimal Rademacher projections of growing norm

Let

$$\Omega_d = \{-1, 1\}^d$$

with uniform probability measure, let $p_d = 2d$, and put

$$X_d = L_{p_d}(\Omega_d).$$

For $1 \le j \le d$, write $r_j(\omega) = \omega_j$ and set

$$E_d = \operatorname{span}\{r_1, \dots, r_d\}.$$

Define the first-chaos, or Rademacher, projection by

$$\mathcal{R}_d f = \sum_{j=1}^{d} \mathbb{E}(f r_j)\, r_j. \tag{4}$$

Lemma 4.1 (Unique minimality)**.** *For every $d \ge 2$, the operator $\mathcal{R}_d : X_d \to E_d$ is the unique minimal projection onto E_d.*

Proof. Identify X_d with the subspace of $L_{p_d}[0,1]$ consisting of functions measurable with respect to the sigma-algebra generated by the first d classical Rademacher functions. Let L_d be conditional expectation onto this finite sigma-algebra; then $\|L_d\| = 1$. Let $\widetilde{\mathcal{R}}_d$ be the classical Rademacher projection on $L_{p_d}[0,1]$, and let T_d be its restriction to X_d. Since the Rademacher functions are measurable with respect to the finite sigma-algebra,

$$\widetilde{\mathcal{R}}_d = T_d L_d.$$

Lewicki and Skrzypek proved that $\widetilde{\mathcal{R}}_d$ is the unique minimal projection onto E_d in $L_p[0,1]$ for every $1 < p < \infty$ [2, 4].

We transfer that statement to X_d. Restriction gives $\|T_d\| \le \|\widetilde{\mathcal{R}}_d\|$, while the displayed factorization and $\|L_d\| = 1$ give the reverse inequality. If $Q : X_d \to E_d$ is any projection, then QL_d is a projection from $L_{p_d}[0,1]$ onto E_d. It follows that

$$\|T_d\| = \lambda(E_d, L_{p_d}[0,1]) \le \|Q\|.$$

Thus T_d is minimal on X_d. If Q is another minimal projection on X_d, then QL_d is minimal on $L_{p_d}[0,1]$ and hence, by uniqueness, $QL_d = \widetilde{\mathcal{R}}_d$. Restriction to X_d gives $Q = T_d$. Under the finite-cube identification, T_d is exactly (4). □

Lemma 4.2 (Norm growth)**.** *For every $d \ge 2$,*

$$\|\mathcal{R}_d\|_{X_d \to X_d} \ge \sqrt{d/6}.$$

Proof. The same linear map acts on every $L_p(\Omega_d)$. Its kernel formula is

$$(\mathcal{R}_d f)(\omega) = 2^{-d} \sum_{\eta \in \Omega_d} f(\eta) \sum_{j=1}^{d} \omega_j \eta_j.$$

Therefore its L_∞ operator norm is the common absolute row sum

$$\|\mathcal{R}_d\|_{\infty\to\infty} = \beta_d := \mathbb{E}\left|\sum_{j=1}^{d} \varepsilon_j\right|,$$

where the ε_j are independent symmetric signs.

On a probability space with N atoms of equal mass,

$$N^{-1/p}\|g\|_\infty \le \|g\|_p \le \|g\|_\infty.$$

Choosing an L_∞ norming vector for the finite matrix of $\mathcal{R}_d$ gives

$$\|\mathcal{R}_d\|_{p\to p} \ge N^{-1/p}\|\mathcal{R}_d\|_{\infty\to\infty}.$$

Here $N = 2^d$ and $p = p_d = 2d$, so

$$\|\mathcal{R}_d\|_{p_d\to p_d} \ge 2^{-1/2}\beta_d. \tag{5}$$

Let $S_d = \sum_{j=1}^{d} \varepsilon_j$. Hölder's inequality gives

$$\mathbb{E}S_d^2 \le (\mathbb{E}|S_d|)^{2/3}(\mathbb{E}S_d^4)^{1/3},$$

and hence

$$\mathbb{E}|S_d| \ge \frac{(\mathbb{E}S_d^2)^{3/2}}{(\mathbb{E}S_d^4)^{1/2}}.$$

Since

$$\mathbb{E}S_d^2 = d, \qquad \mathbb{E}S_d^4 = 3d^2 - 2d \le 3d^2,$$

we obtain $\beta_d \ge \sqrt{d/3}$. Combining this with (5) proves the lemma. □

Proof of Theorem 1.1. Apply Proposition 3.2 to the unique minimal projection $\mathcal{R}_d : X_d \to E_d$. It supplies a hyperplane F_d containing E_d such that $\lambda(F_d, X_d) \le 2$. Set

$$X_{1,d} = E_d, \qquad X_{2,d} = F_d, \qquad X_{3,d} = X_d.$$

If $P_1 : F_d \to E_d$ and $P_2 : X_d \to F_d$ are projections with $\|P_1P_2\| = \lambda(E_d, X_d)$, then P_1P_2 is a minimal projection onto E_d. Lemma 4.1 makes it equal to $\mathcal{R}_d$, and Proposition 3.2, followed by Lemma 4.2, gives

$$\|P_2\| \ge \|\mathcal{R}_d\| - 2 \ge \sqrt{d/6} - 2.$$

Since $\lambda(F_d, X_d) \le 2$, the ratio in the theorem tends to infinity. □

5 Consequences and literature status

Define C_d^* to be the least constant that works uniformly in Problem 3 over all triples with $\dim X_1 = d$. Corollary 2.2 and Theorem 1.1 show

$$\frac{\sqrt{d/6} - 2}{2} \le C_d^* \le 1 + 2\sqrt{d}.$$

Thus the $\sqrt{d}$ dependence in the supplied partial result is optimal in order, although the numerical constants have not been optimized.

A targeted search of the title and exact wording of Ostrovskii's Problem 3, together with searches through visible citation trails and projection-constant literature, did not locate a paper explicitly giving a full answer or counterexample. This is not an exhaustive priority claim. The construction above uses a published result that predates the problem: the unique minimality of Rademacher projections proved by Lewicki and Skrzypek and restated explicitly by Skrzypek. All other ingredients in the counterexample are included here.

Human review notes

The finite-rank correction formula in Proposition 2.1 is elementary and is retained because it gives the matching upper estimate. The main mathematical review point for the negative answer is Lemma 4.1: one should check the cited unique minimality theorem for the classical Rademacher projection and the conditional-expectation transfer from $L_{2d}[0,1]$ to the finite cube. Once that input is accepted, the hyperplane obstruction and norm-growth estimates are finite-dimensional and self-contained.

References

[1] M. I. Kadets and M. G. Snobar, *Certain functionals on the Minkowski compactum*, Math. Notes **10** (1971), 694–696.

[2] G. Lewicki and L. Skrzypek, *Chalmers–Metcalf operator and uniqueness of minimal projections*, J. Approx. Theory **148** (2007), no. 1, 71–91. doi:10.1016/j.jat.2006.10.008.

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[4] L. Skrzypek, *On the L_p norm of the Rademacher projection and related inequalities*, Proc. Amer. Math. Soc. **137** (2009), no. 8, 2661–2669. doi:10.1090/S0002-9939-09-09875-X.